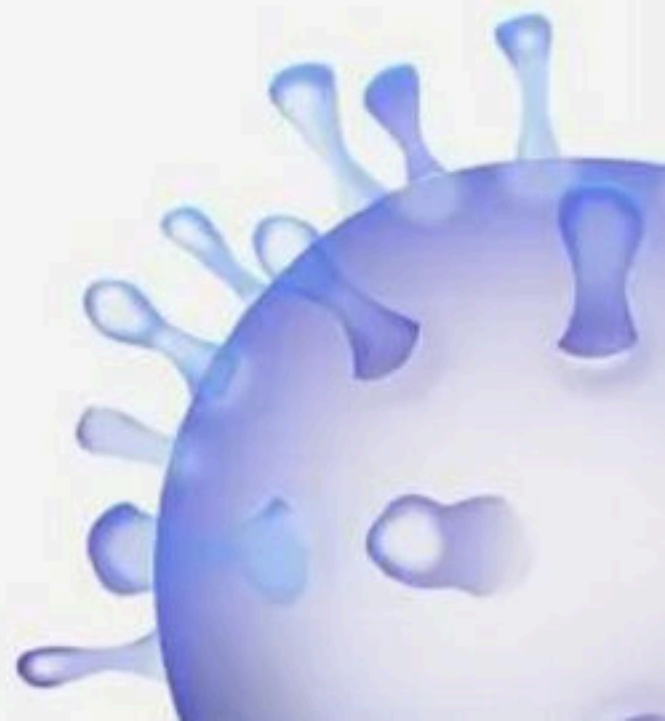


Evolution - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Evolution



BIOLOGYLIVE

1. Evolutionary history of an organism is also known as

- (1) Phylogeny
- (2) Paleontology
- (3) Ontogeny
- (4) Phycology



2. According to Lamarckism, long necked Giraffes evolved because

- (1) Humans preferred only long necked ones
- (2) Short necks suddenly changed into long necks
- (3) Nature selected only long necked ones
- (4) Stretching of necks over many generation by short necked ones



3. In a hypothetical human population in HardyWeinberg equilibrium, the frequencies of the alleles I^A , I^B and i are respectively 0.1, 0.3 and 0.6 . What percentage of the population is expected to have 'B' type blood?

- (1) 27
- (2) 9
- (3) 36
- (4) 45



4. Which of the following are the two key concepts of Darwinian theory of evolution?

- (1) Branching descent and mutation
- (2) Adaptive radiation and branching ascent
- (3) Mutation and natural selection
- (4) Branching descent and natural selection



5. Given below are some stages of human evolution. Arrange them in correct sequence. (Past to Recent)

A. Homo habilis

B. Homo sapiens

C. Homo neanderthalensis

D. Homo erectus

Choose the correct sequence of human evolution from the options given below:

(1) D - A - C - B

(2) B - A - D - C

(3) C - B - D - A

(4) A - D - C - B



6. Evolution of man took place in

- (1) Australia
- (2) Africa
- (3) Central America
- (4) Central Asia



7. If the gene flow occurs by chance, it is called?

- (1) Genetic shift
- (2) Genetic drift
- (3) Genetic migration
- (4) None of them



8. In which respect does Darwin's theory is wrong?

- (1) Survival of the fittest
- (2) Arrival of the fittest
- (3) High efficiency of reproduction
- (4) Origin of species



9. "Given below are four statements (A - D) each with one or two blanks. Select the option which correctly fills up the blanks in two statements.

(A) Wings of butterfly and birds look alike and are the results of ____ (i) ____ volution.

(B) Miller showed that CH_4 , H_2 , NH_3 and ____ (i) ____ when exposed to electric discharge in a flask resulted in formation of ____ (ii) ____.

(C) Vermiform appendix is a ____ (i) ____ organ and an ____ (ii) ____ evidence of evolution.

(D) According to Darwin evolution took place due to ____ (i) ____ and ____ (ii) ____ of the fittest.

Options:

(1) (A) - (i) convergent,

(B) - (i) Oxygen, (ii) nucleosides.

(2) (B) - (i) Water vapour, (ii) Amino acids,

(C) - (i) vestigial, (ii) Anatomical.

(3) (C) - (i) vestigial, (ii) anatomical,

(D) - (i) Mutations, (ii) Multiplication.

(4) (D) - (i) small variations, (ii) survival,

(C) - (i) convergent, (ii) adaptive



10. Which is amongst the following show common ancestry line of mammals?

- (1) Therapsids
- (2) Synapsids
- (3) Pelycosaurs
- (4) All of these



11. Fill in the blanks :

Mutations are _____ while Darwinian variations are _____

- (1) Constant and directionless, small and directional.
- (2) Random and directional, small and directional.
- (3) Random and directionless, small and directional.
- (4) Random and directionless, small and directionless



12. Who proved the theory of chemical evolution through his experiment?

- (1) L.S. Miller, 1953
- (2) S.L. Miller, 1952
- (3) S.L. Miller, 1953
- (4) L.S. Miller, 1952



13. Statement A : Ernst Haeckel supported the embryological similarities as an evidence of evolution.

Statement B : The embryos of all vertebrates including human develop a row of vestigial gill slits which are absent in adult vertebrates.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



14. Which of the following statement is false?

- (1) Microbial experiments show that pre-existing advantageous mutations when selected will result in observation of new phenotypes. Over few generations, this would result in Speciation.
- (2) The sum total of genes of all the species in an area constitutes the gene pool.
- (3) The gene frequency refers to the proportion of an allele in the gene pool as compared with other alleles.
- (4) None of the above



15. The theory of natural selection given by Charles Darwin was further supported by

- (1) Lamarck
- (2) Alfred Wallace
- (3) Darwin
- (4) Oparian and Haldane



16. Random genetic drift in a population probably results from

- (1) Large population size
- (2) Highly genetically variable individuals
- (3) Constant low mutation rate
- (4) Interbreeding within this population



17. Among the following sets of examples for divergent evolution, select the incorrect option:

- (1) Brain of bat, man and cheetah
- (2) Heart of bat, man and cheetah
- (3) Forelimbs of man, bat and cheetah
- (4) Eye of octopus, bat and man



18. Which of the following is used as an atmospheric pollution indicator

- (1) Lepidoptera
- (2) Lichens
- (3) Lycopersicon
- (4) Lycopodium



19. In a population of plants, some were extremely tall and the remaining were extremely dwarf. No plants of the population showed intermediate height. The type of operation of natural selection in the above case is

- (1) Balancing
- (2) Directional
- (3) Stabilizing
- (4) Disruptive



20. A potential damage to a population that has been greatly reduced in number is the

- (1) Loss of genetic variability
- (2) Tendency towards assortative mating
- (3) Reduced gene flow
- (4) Hardy - Weinberg disequilibrium



21. The gene pool represents _____ in a population and it remains a constant.

- (1) Total genes only
- (2) Dominant alleles only
- (3) Recessive alleles only
- (4) Total genes and their alleles



22. The change of the lighter coloured variety of peppered moth, *Biston betularia* to its darker variety (*carbonaria*) is due to

- (1) Deletion of a segment of genes due to industrial pollution
- (2) Translocation of a block of genes in chromosomes in response to heavy carbons
- (3) Industrial carbon deposited on the barks of the trees resulting in darker variety being more camouflaged
- (4) Mutation of single mendelian gene for survival in smoke laden industrial environment.



23. Statement A : The skull of adult chimpanzee is more like adult human skull than baby chimpanzee skull.

Statement B : The skull of baby chimpanzee is more like adult human than adult chimpanzee skull.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



24. Statement A : Giant ferns fell to form coal deposits slowly.

Statement B : About 65 million years ago dinosaurs died out.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



25. When more than one adaptive radiation appeared to have occurred in an isolated geographical area, it is

- (1) Convergent evolution
- (2) Divergent evolution
- (3) Natural selection
- (4) Genetic drift



26. Which of the following structures are 'homologous' from the evolutionary point of view?

- (1) Petals of rose and the anther of an apple blossom.
- (2) Thorn of bougainvillea and tendrils of cucurbita
- (3) Spines of a cactus and spines of a porcupine.
- (4) Flagellum of Euglena and the flagella of Paramecium



27. Dinosaurs became extinct during ____ time period.

- (1) Jurassic period
- (2) Triassic period
- (3) Permian Period
- (4) Cretaceous period



28. The eye of octopus and eye of cat show different patterns of structure, yet they perform similar function. This is an example of

- (1) Homologous organs that have evolved due to divergent evolution.
- (2) Analogous organs that have evolved due to convergent evolution.
- (3) Analogous organs that have evolved due to divergent evolution.
- (4) Homologous organs that have evolved due to convergent evolution.



29. On critical analysis which of the following has resulted in changed frequency of genes and alleles in future generation?

- (1) Variation due to mutation
- (2) Variation due to recombination
- (3) Gene flow or genetic drift
- (4) All of these



30. Who said that evolution of the life forms has occurred but driven by use and disuse of organs:

- (1) Darwin - French naturalist
- (2) Lamarck - French naturalist
- (3) De Vries - French naturalist
- (4) Malthus - French naturalist



31. In a population that is in Hardy-Weinberg equilibrium, the frequency of the recessive homozygote genotype of a certain trait is 0.09 . Percentage of individuals homozygous for the dominant allele is

- (1) 0.09
- (2) 0.30
- (3) 0.70
- (4) 0.49



32. Tasmanian Wolf is a marsupial while Wolf is a placental mammal. This is an example of

- (1) Convergent evolution
- (2) Divergent evolution
- (3) Inheritance of acquired characters
- (4) Adaptive radiation



33. The process by which different type of finches were evolved in Galapagos islands is a result of

- (1) Adaptive radiation
- (2) Non -adaptive radiation
- (3) Industrial melanism
- (4) Convergent evolution



34. Hardy-Weinberg law is applicable to

- (1) Small population
- (2) Genetic equilibrium
- (3) Genetic drift
- (4) Non-random mating



35. S.L. Miller created electric discharge in a closed flask containing.

- (1) CH_4 , H_2 , NH_3 and water vapour at 80°C
- (2) CH_4 , H_2 , NH_3 and water vapour at 800°C
- (3) CH_4 , H_2 , NH_3 and water vapour at 8000°C
- (4) CH_4 , H_2 , NH_2 and water vapour at 800°C



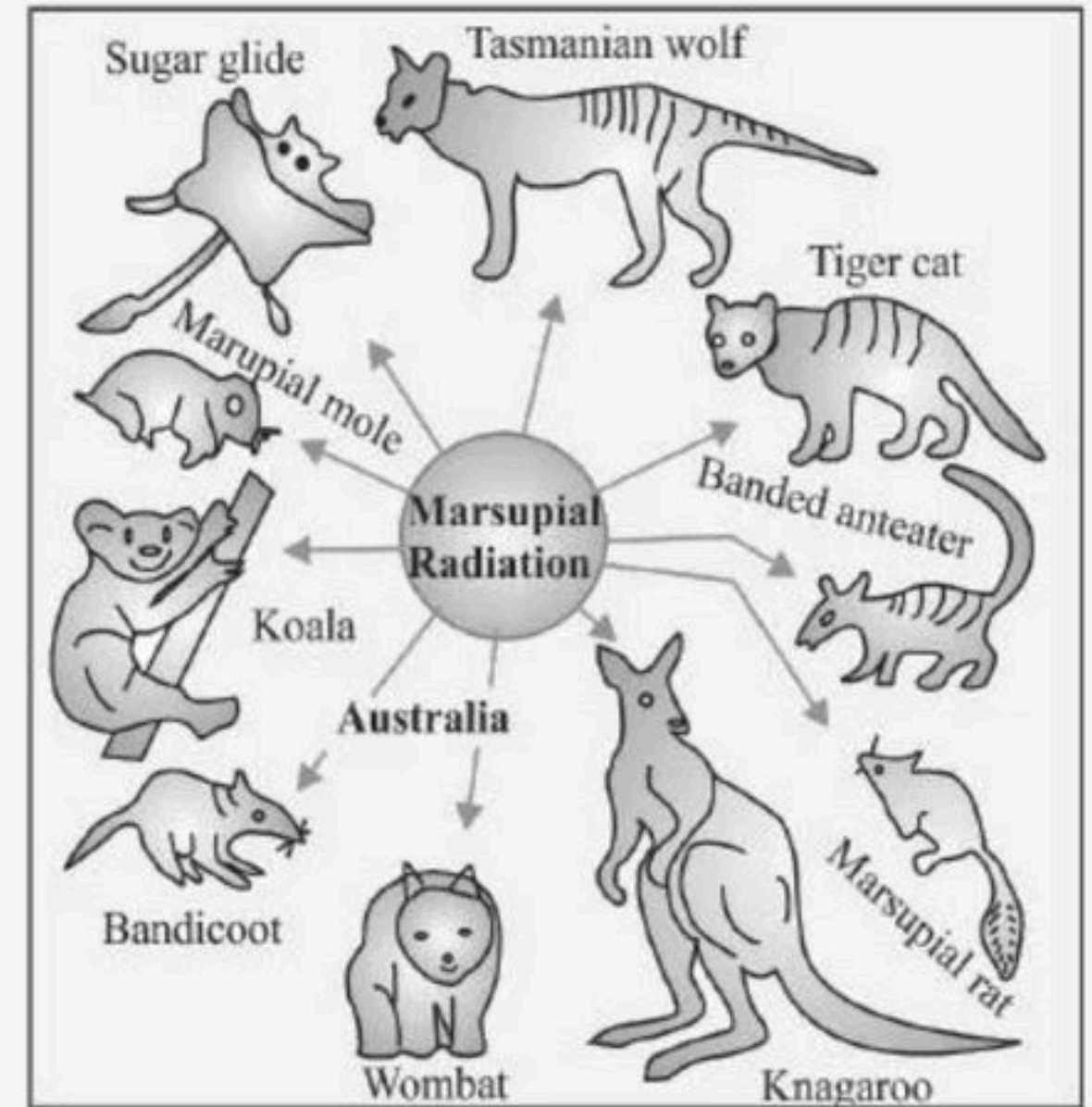
36. Variations during mutations of meiotic recombinations are

- (1) Random and directionless
- (2) Random and directional
- (3) Small and directional
- (4) Random, small and directional



37. The given figure represents

- (1) Adaptive Radiation of marsupials of Australia
- (2) Convergent evolution of Homo erectus
- (3) Divergent evolution of Marsupials of Australia
- (4) Family tree of Homo erectus



38. Divergent evolution gives rise to

- (1) Homologous organs
- (2) Analogous organs
- (3) Both (1) and (2)
- (4) None of these



39. Statement A : Sudden inheritable changes take place in genome of an organism due to certain factors called mutations.

Statement B : Lamarck's theory of evolution is popularly called the theory of continuity of germplasm.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



40. Example of convergent evolution is

- (1) Darwin finches and marsupial mouse
- (2) Placental wolf and Tasmanian wolf
- (3) Lemur and Spotted cuscus
- (4) Both 2 and 3



41. Millers experiment provided evidence for theory of

- (1) Biogenesis
- (2) Abiogenesis
- (3) Special creation
- (4) Organic evolution



42. Analogous structures are a result of

- (1) Convergent evolution
- (2) Shared ancestry
- (3) Stabilizing selection
- (4) Divergent evolution



43. Match the following columns.

Column I		Column II	
A	Stabilisation	P	Industrial melanism
B	Directional selection	Q	More individuals possess heterozygous character
C	Disruption selection	R	More individuals possess homozygous character

- (1) A – R, B – P, C – Q
(2) A – Q, B – P, C – R
(3) A – P, B – R, C – Q
(4) A – Q, B – R, C – P



44. Match the column I with Column II

Column I

A Mesozoic

B Devonian

C Palaeocene

D Permian

Column II

P First amphibians

Q 300-250 mya

R 160 million years

S Radiation of primitive mammals

(1) A – R, B – P, C – R, D – Q

(2) A – R, B – P, C – S, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



45. Match the following columns.

Column I		Column II	
A	Genetic drift	P	Change in the population's allele frequency due to chance alone
B	Natural selection	Q	Selection of better traits
C	Gene flow	R	Immigration or emigration changes the allele frequency
D	Mutation	S	Saltation

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – Q, C – R, D – S

(4) A – Q, B – P, C – R, D – S



46. Match the scientists listed under column 'I' with ideas listed Column 'II'

Column I		Column II	
A	Darwin	P	Abiogenesis
B	Oparin	Q	Use and disuse of organs
C	Lamarck	R	Germ plasm theory and neodarwinism
D	August Weisman	S	Evolution by natural selection

(1) A - P, B - S, C - Q, D - R

(2) A - S, B - P, C - Q, D - R

(3) A - Q, B - S, C - R, D - P

(4) A - S, B - R, C - Q, D - P



47. Match Column I with Column II:

Choose the correct answer from the options given below :

Column I

A Mesozoic Era

B Proterozoic Era

C Cenozoic Era

D Paleozoic Era

Column II

P Lower invertebrates

Q Fish & Amphibians

R Birds & Reptiles

S Mammals

(1) A – Q, B – P, C – R, D – S

(2) A – R, B – P, C – Q, D – S

(3) A – P, B – Q, C – S, D – R

(4) A – R, B – P, C – S, D – Q



48. Match Column - I with Column - II.

Column I		Column II	
A	Adaptive radiation	P	Selection of resonance varieties due to excessive use of herbicides and pesticides
B	Convergent evolution	Q	Bones of forelimbs in man and whale
C	Divergent evolution	R	Wings of butterfly and bird
D	Evolution by anthropogenic action	S	Darwin finches

(1) A - S; B - R; C - Q; D - P

(2) A - R; B - Q; C - P; D - S

(3) A - Q; B - P; C - S; D - R

(4) A - P; B - S; C - R; D - Q



49. Match the columns I and II.

Column I

Column II

A Oparin

P America

B J.B.S. Haldane

Q Russia

C Miller

R England

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



50. Match the following columns.

Column I

A Wallace

B Malthus

C Hardy-Weinberg Law

D Industrial melanism

Column II

P Archipelago

Q Influenced Darwin

R England

S $p^2 + 2pq + q^2 = 1$

(1) A – R, B – P, C – Q, D – S

(3) A – P, B – R, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(4) A – P, B – Q, C – S, D – R



51. Match the following columns.

Column I		Column II	
A	Darwin	P	Branching descent
B	Lamarck	Q	Inheritance of aquired character
C	Pasteur	R	Disapproved spontaneous theory
D	De Vries	S	Mutational theory of inheritance

(1) A – R, B – P, C – Q, D – S

(2) A – P, B – Q, C – R, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



52. Match the columns I and II.

Column I		Column II	
A	<i>Homo habilis</i>	P	900 cc
B	<i>Homo neanderthalensis</i>	Q	650-800 cc
C	<i>Homo erectus</i>	R	1400 cc
D	<i>Homo sapiens</i>	S	1350 cc

- (1) A – R, B – P, C – Q, D – S
- (2) A – S, B – R, C – P, D – Q
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – R, C – P, D – S



53. Identify the correct match from column-I and column- II:

Column I

A. Origin of universe

B. Origin of Earth

C. Origin of first cellular form of life

D. Origin of first noncellular form of life

Column II

P. 3 bya

Q. 20 bya

R. 4.5 bya

S. 2 bya

(1) A - P, B - Q, C - R, D - S

(3) A - Q, B - S, C - R, D - P

(2) A - Q, B - R, C - S, D - P

(4) A - P, B - R, C - Q, D - S



54. Match the columns I and II.

Column I		Column II	
A	Pasteur	P	Life only comes from life
B	Oparin	Q	Experimental verification of chemical evolution
C	S.L. Miller	R	Theory of chemical evolution

- (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



55. Match the following columns.

Column I		Column II	
A	<i>Ichthyosaurus</i>	P	Caught in South Africa in 1938
B	Coelacanth	Q	Fell to form coal deposits
C	Giant pteridophytes	R	Disappeared about 65 mya
D	Dinosaurs	S	Fish-like reptiles in 200 mya

- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – S, B – P, C – Q, D – R



56. Match the following

Column I		Column II	
A	First to use fire	P	<i>Homo erectus</i>
B	Ape like primate	Q	<i>Dryopithecus</i>
C	Ancestor of modern apes	R	<i>Australopithecus</i>
D	Connecting link between ape and man	S	<i>Propliopethecus</i>

- (1) A – P, B – S, C – Q, D – R
(2) A – Q, B – R, C – P, D – S
(3) A – Q, B – S, C – R, D – P
(4) A – R, B – P, C – S, D – Q



57. Statement A : Wings of insects and bats are analogous.

Statement B : Wings of insects and birds are analogous.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



58. Read the statements A and statement B carefully to mark the correct options given below.

Statement A : All vertebrates develop a row of vestigial gill slits during embryonic stage.

Statement B : Embryos always pass through the adult stages of other animals.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



59. Which of the following statements regarding the evolution of plants and animals are correct?

I. Reptiles evolved into birds

II. Fish with stout and strong fins could move on land and go back to water. This was about 350 million years ago.

III. Giant ferns fell to form coal deposits slowly.

IV. About 65 billion years ago dinosaurs died out.

V. Archaeopteryx is the connecting link between birds and reptiles.

The correct answer is

(1) I and II

(2) III and IV

(3) V and I

(4) I, II, III, & V



60. Statement A : In natural selection, heritable variations enable better survival

Statement B : Australopithecus hunted with stone weapons

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



61. Statement A : Charles Darwin was on a sea voyage in a sail ship called H.M.S. Beagle

Statement B : Charles Darwin concluded that existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



62. Statement A : Natural selection is an outcome of the differences in survival and reproduction among individuals that show variations in one or more traits.

Statement B : Adaptive forms of a given trait tend to become more common.

Less adaptive ones beomes less common or disappear.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



63. Consider the following statements.

I. Homo habilis: The brain capacities were between 650-800cc.

II. Erect man or Homo erectus skull was flatter than that of modern man.

III. Homo erectus probably ate meat

IV. Neanderthal man is regarded as most recent ancestor of today's man.

Choose the correct statements.

(1) I, III and IV

(2) I, II and III

(3) I and II

(4) I and III



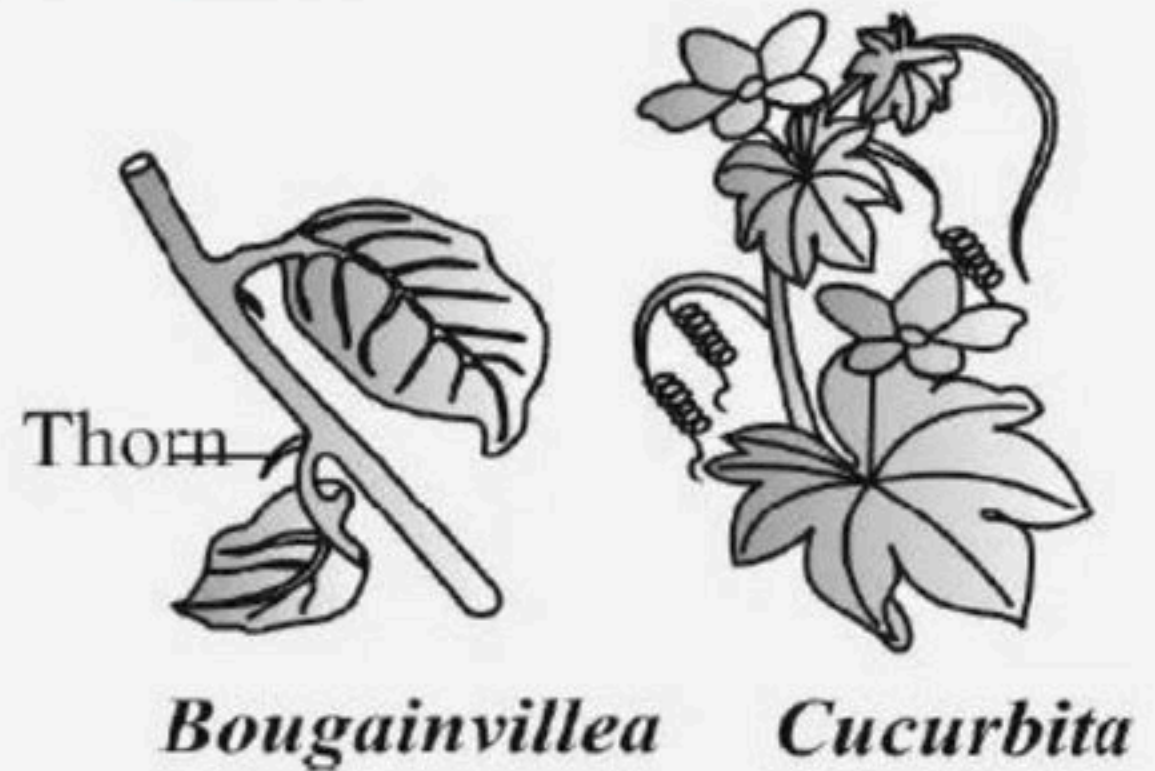
64. Oparin of _____ and Haldane of _____ proposed that the first form of life could have come from preexisting non-living organic molecules.

- (1) Russia, Russia
- (2) Russia, England
- (3) England, England
- (4) England, Russia



65. The following diagram represents homologous organ, which part of the plant represents homology?

- (1) Root tendrils
- (2) Leaf tendrils
- (3) Stem tendrils
- (4) Spine tendrils



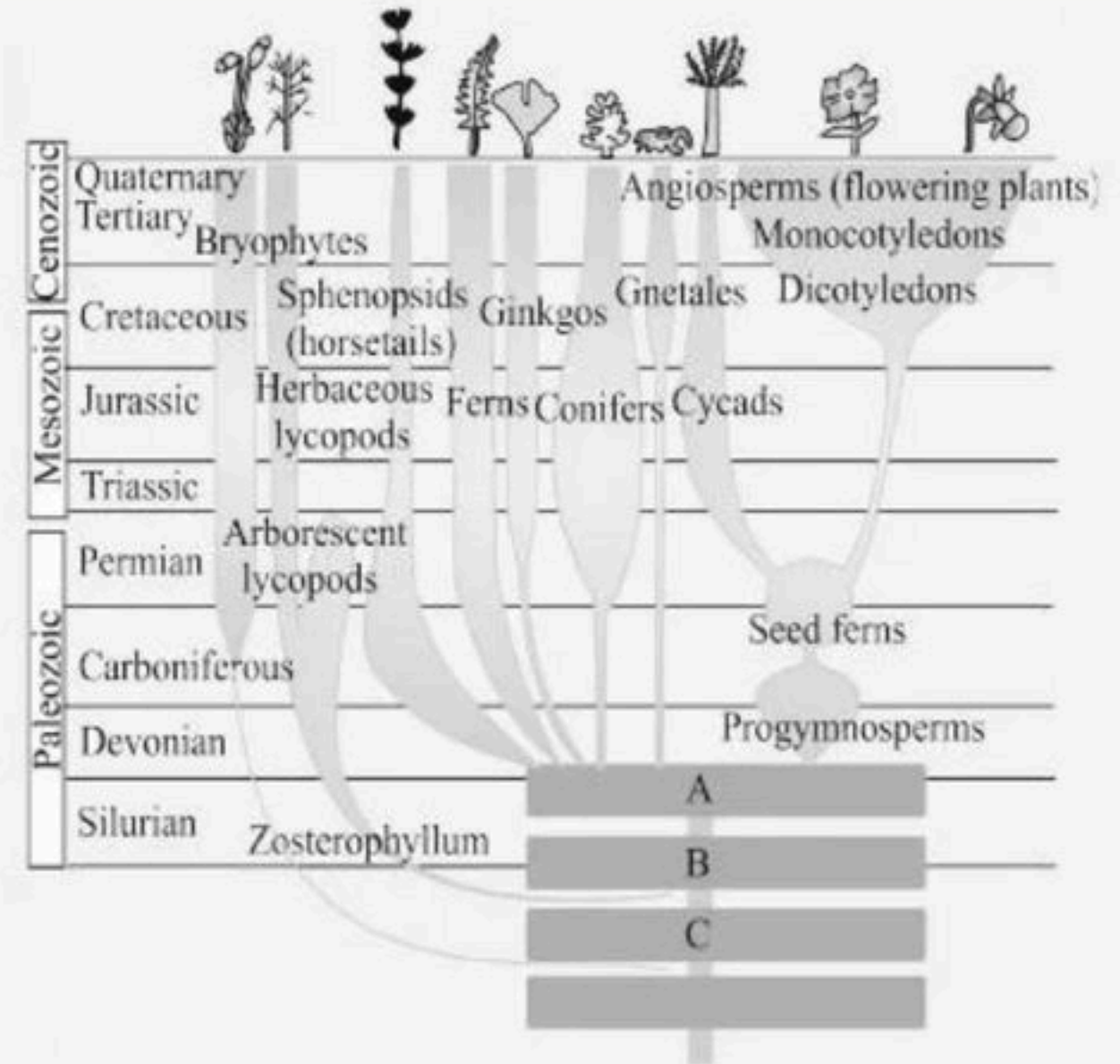
66. Identify the missing ancestors.

(1) A - Chlorophyte ancestor, B - Rhynia type, C - Tracheophyte

(2) A - Psilophyton, B - Rhynia type, C - Tracheophyte

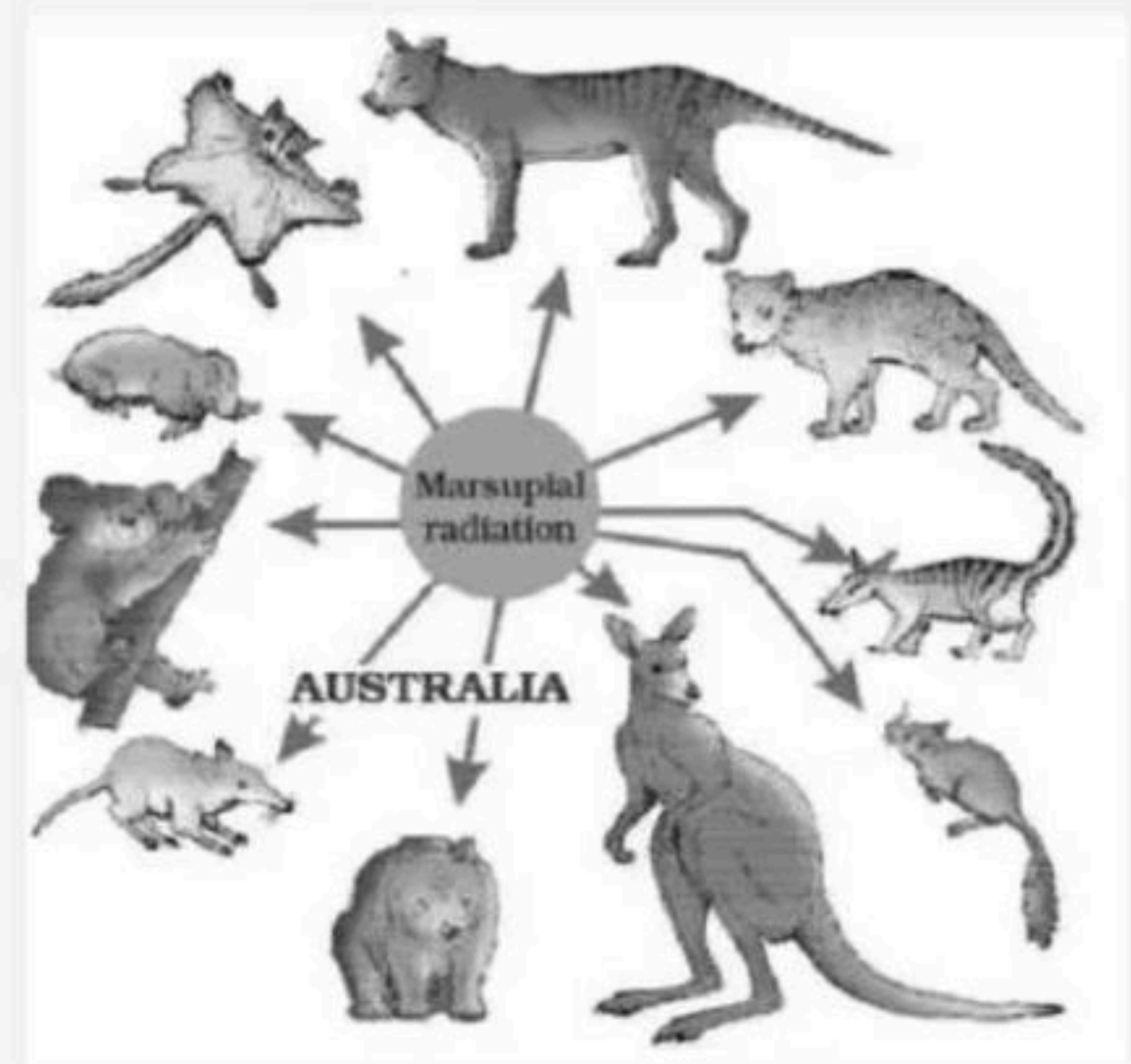
(3) A - Psilophyton, B - Tracheophyte, C - Chlorophyte

(4) A - Psilophyton, B - Rhynia, C - Chlorophyte



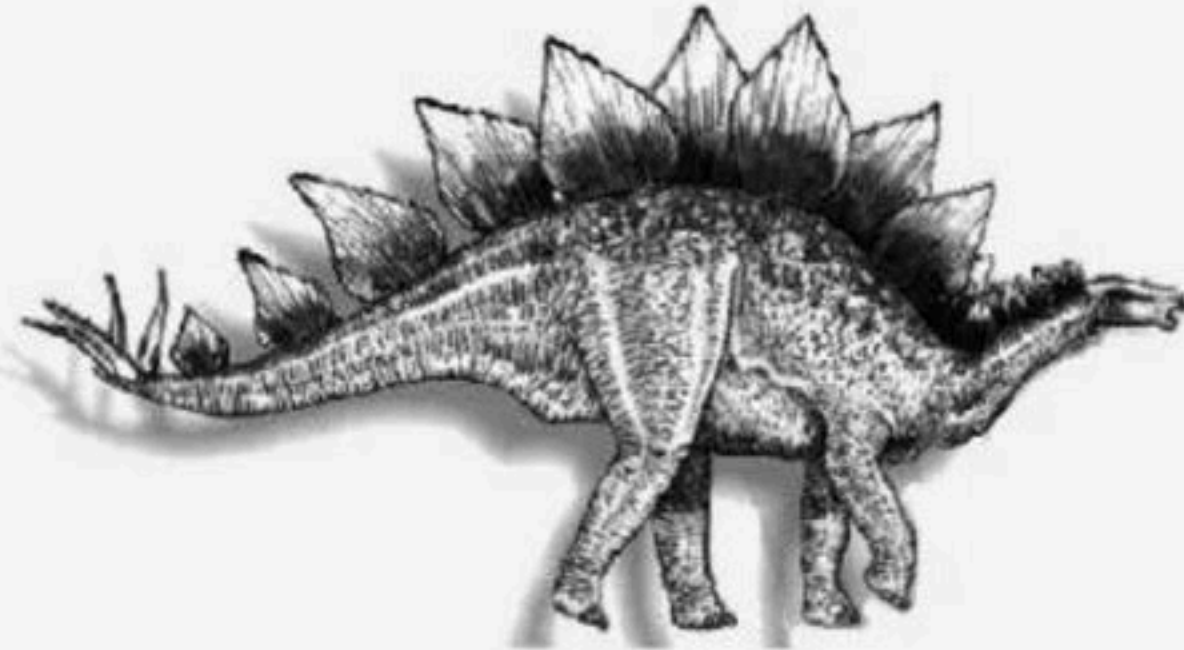
67. What does the following diagram represent?

- (1) Natural selection
- (2) Marsupial radiation
- (3) Placental radiation
- (4) Mutation



68. Identify the following dinosaur.

- (1) Branchiosaurus
- (2) Stegosaurus
- (3) Archeopteryx
- (4) Triceratops



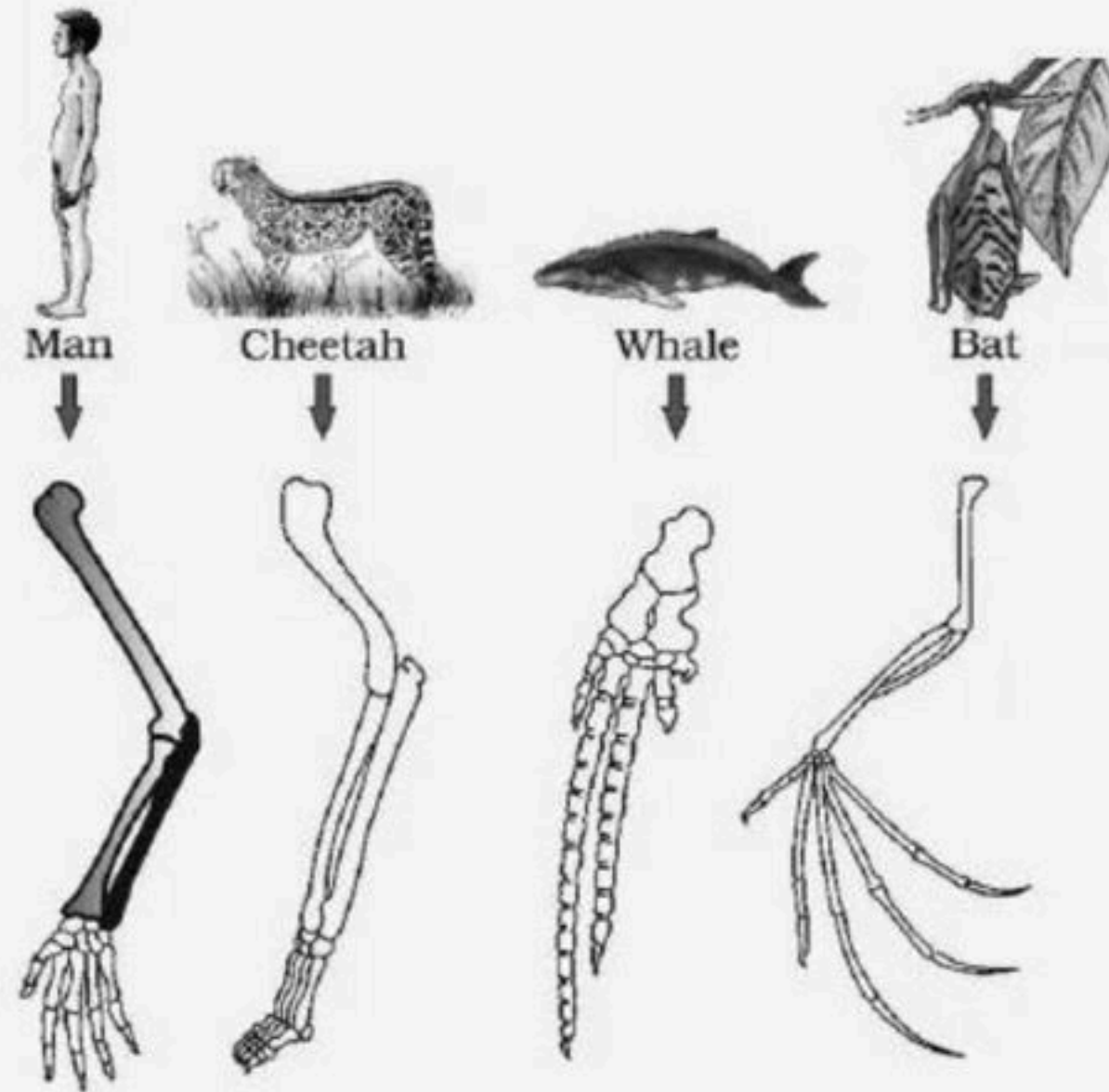
69. The flowering insect represents a particular phenomenon in England. Identify the correct option.

- (1) Mutation
- (2) Natural selection
- (3) Adaptive radiation
- (4) Convergent evolution



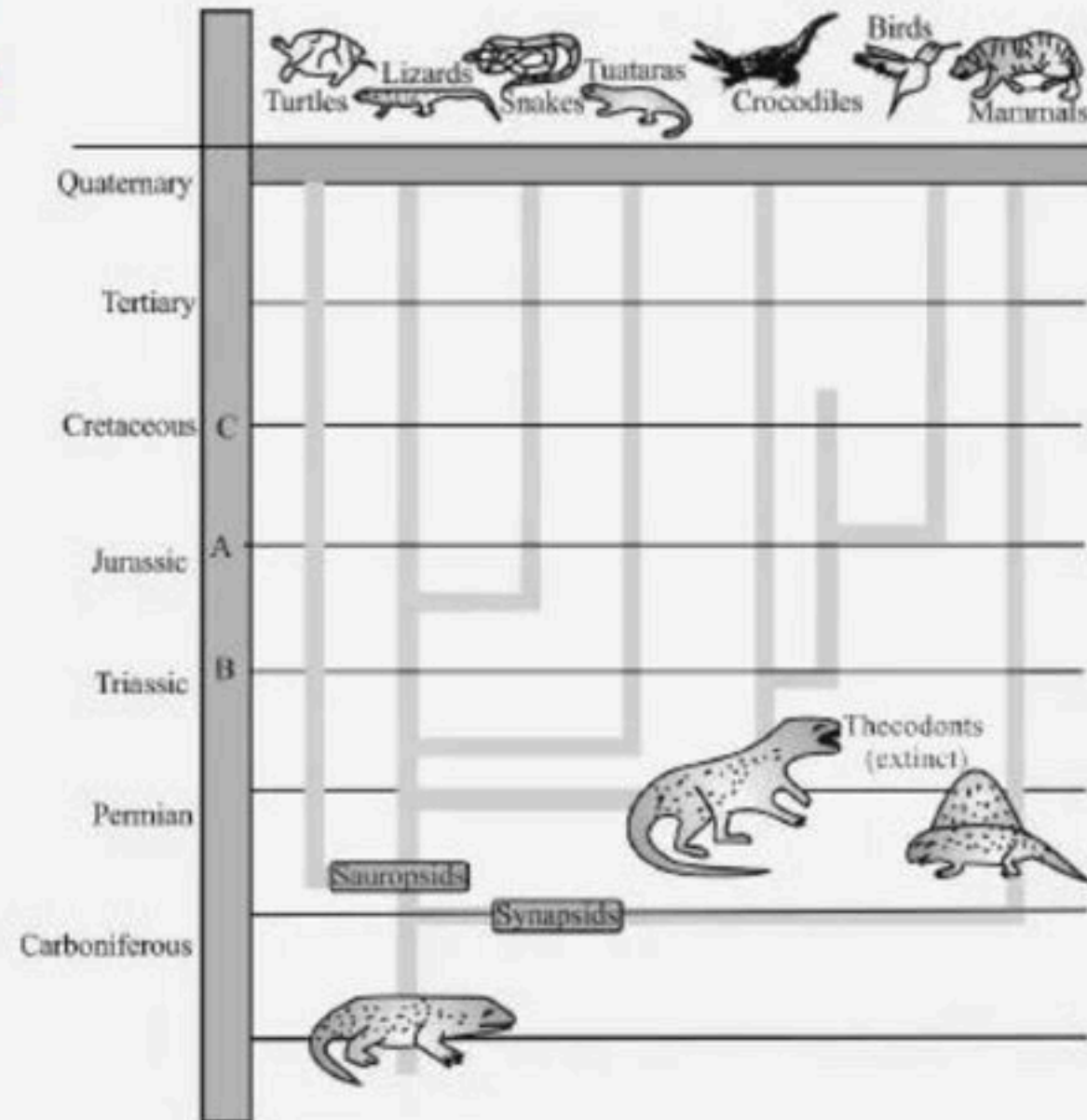
70. The following diagram represents

- (1) Homology in plants
- (2) Homology in animals
- (3) Divergent evolution
- (4) Both 2 & 3



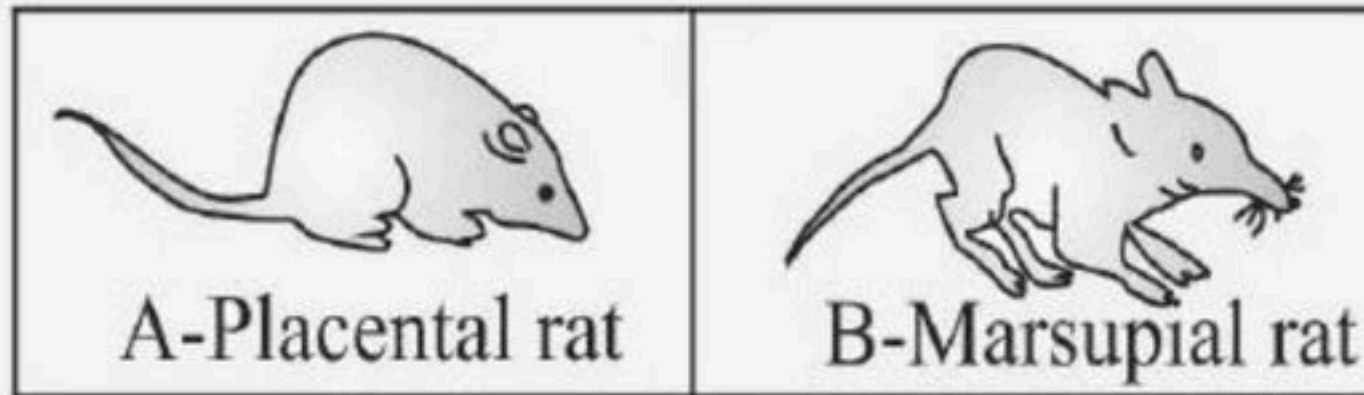
71. Identify the missing time period.

- (1) A - 150, B - 100, C - 200
- (2) A - 150, B - 200, C - 100
- (3) A - 300, B - 350, C - 250
- (4) A - 250, B - 350, C - 300



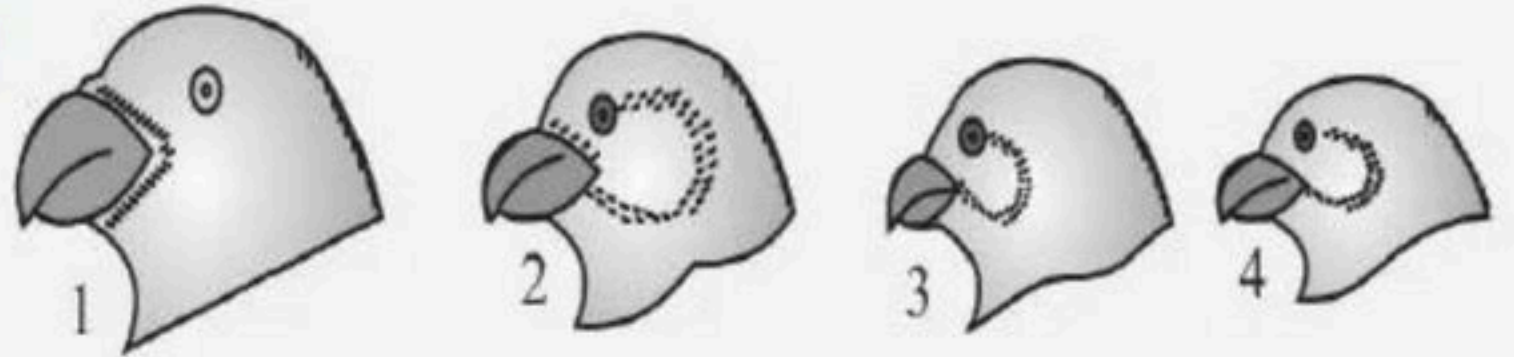
72. The organism A and B are example of:

- (1) Convergent evolution
- (2) Divergent evolution
- (3) Adaptive convergence
- (4) Both (1) and (3)



73. What does the given image depicts?

- (1) Variety of beaks of Darwins finches
- (2) Variety of birds of African island
- (3) Variety of beaks of finches found in Galapagos island.
- (4) Both (1) and (3)



74. The given figure shows an example of

- (1) Homologous organs
- (2) Convergent evolution
- (3) Analogous organs
- (4) Vestigial organs



75. The given figure represents

- (1) Divergent evolution
- (2) Adaptive radiation
- (3) Convergent evolution
- (4) Classification

Niche	Placental Mammals	Australian Marsupials
Burrower	Mole 	Marsupial mole 
Anteater	Anteater 	Numbat(anteater) 
Mouse	Mouse 	Marsupial mouse 
Climber	Lemur 	Spotted cuscus 
Glider	Flying squirrel 	Flying phalanger 
Cat	Bobcat 	Tasmanian "tiger cat" 
Wolf	Wolf 	Tasmanian Wolf 

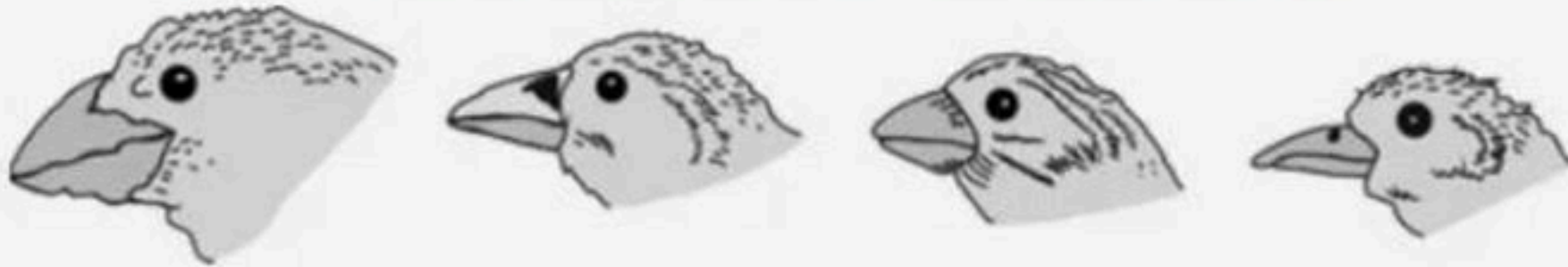


76. The given diagram of marsupials of Australia provides an example of

- (1) Convergent evolution
- (2) Parallel evolution
- (3) Recapitulation
- (4) Divergent evolution



77. There is a difference in one morphological character in the birds. The part involved in the change is



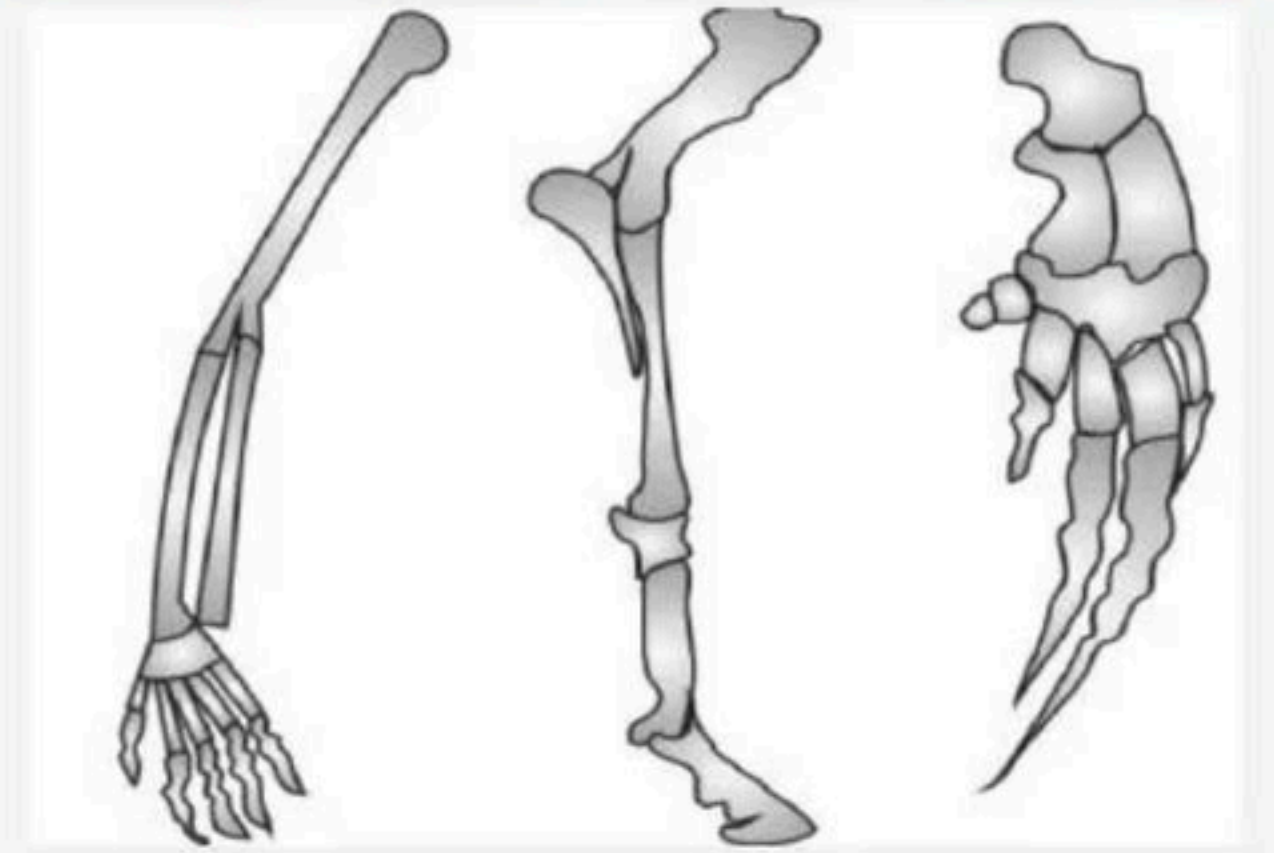
- (1) Size of eye
- (2) Shape of beak
- (3) Size of bird size/
- (4) Size of wings.



78. The given bones in the forelimbs of three mammals figure shows.

For these mammals, the number, position, and shape of the bones most likely indicates that they may have

- (1) Developed in a common environment.
- (2) Developed from the same earlier species.
- (3) Identical genotype
- (4) Identical methods of obtaining food.



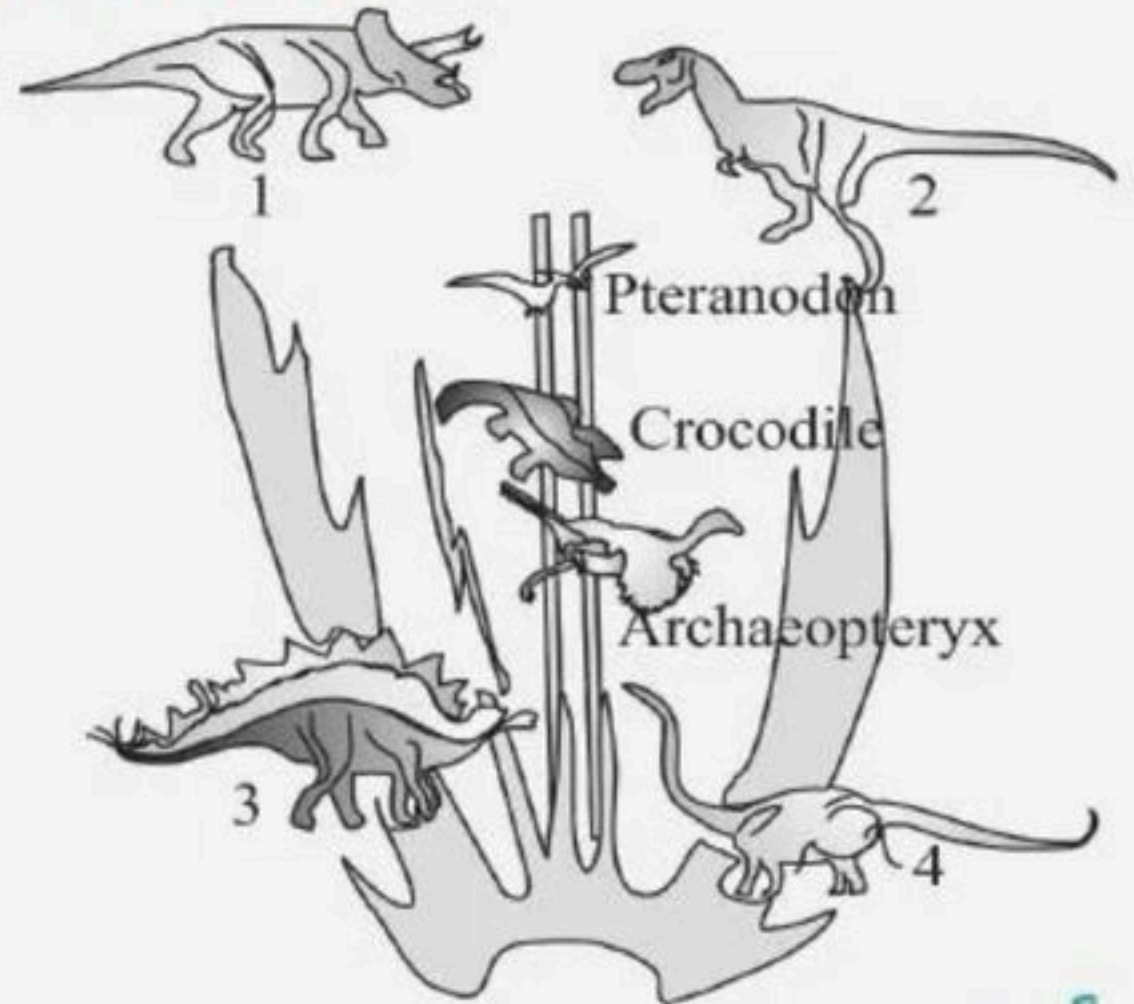
79. A family tree of dinosaurs and their living modern day counterpart organisms like crocodiles and birds are given below. Identify 1,2, 3 and 4.

(1) 1 - Stegosaurus, 2 - Brachiosaurus, 3 - Tyrannosaurus, 4 - Triceratops

(2) 1 - Triceratops, 2 - Tyrannosaurus, 3 - Stegosaurus, 4 - Brachiosaurus

(3) 1 - Stegosaurus, 2 - Triceratops, 3 - Brachiosaurus, 4 - Tyrannosaurus

(4) 1 - Triceratops, 2 - Brachiosaurus, 3 - Stegosaurus, 4 - Tyrannosaurus



80. Which of the following is the correct sequence of events in the origin of life.

- i. Formation of protobionts**
- ii. Synthesis of organic monomers.**
- iii. Synthesis of organic polymers.**
- iv. Formation of DNA- based genetic systems.**

(1) i, ii, iii, iv

(2) i, iii, ii, iv

(3) ii, iii, i, iv

(4) ii, iii, iv, i



81. Forelimbs of cat, lizard used in walking; forelimbs of whale used in swimming and forelimbs of bats used in flying are an example of:

- (1) Analogous organs
- (2) Adaptive radiation
- (3) Homologous organs
- (4) Convergent evolution



82. What was the most significant trend in the evolution of modern man (*Homo sapiens*) from his ancestors?

- (1) Binocular vision
- (2) Upright posture
- (3) Shortening of jaws
- (4) Increasing cranial capacity



83. The flippers of the Penguins and Dolphins are the example of the

- (1) Adaptive radiation
- (2) Natural selection
- (3) Convergent evolution
- (4) Divergent evolution



84. Study the following table and identify the correct option for A, B and C.

	Fossil	Cranial capacity	Time of appearance
I	Australopithecus	A	5 mya
II	Homo sapiens neanderthalensis	1400 cc	B
III	C	900 cc	1.5 mya

- (1) A - 700cc, B - 3400 years ago, C - Homo erectus
- (2) A - 500cc, B - 11000 years ago, C - Homo habilis
- (3) A - 800cc, B - 50,000 years ago, C - Ramapithecus
- (4) A - 500cc, B - 100,000 years ago, C - Homo erectus



85. The ancestors of modern day frogs and Salamanders are

- (1) Jawless fish
- (2) Coelocanth
- (3) Ichthyophis
- (4) Amphioxus



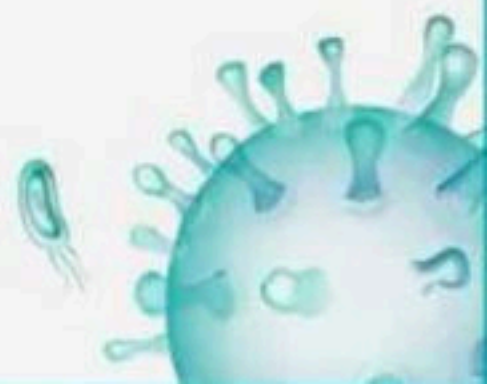
86. Hardy - Weinberg principle is expressed by following equation:

(1) $p^2 - q^2$

(2) $p^2 + q^2$

(3) $p^2 + 2pq + q^2$

(4) $p^2 - 2pq - q^2$



87. Select the correct order of geological time scale of earth.

- (1) Palaeozoic → Archaeozoic → Cenozoic
- (2) Archaeozoic → Palaeozoic → Proterozoic
- (3) Palaeozoic → Mesozoic → Cenozoic
- (4) Mesozoic → Archaeozoic → Proterozoic



**88. The first non-cellular form of life could have originated
_____ billion years back.**

(1) 4.5

(2) 2

(3) 3

(4) 4



89. The correct sequence of evolution of man:

I. Homo habilis

II. Java Man

III. Modern Man

IV. Dryopithecus

(1) II, I, III, IV

(2) IV, I, II, III

(3) I, II, III, IV

(4) I, II, IV, III



90. In a population of 1000 individuals 360 belong to genotype AA, 480 to and the remaining 160 to aa. Based on this data, the frequency of allele A in the population is:

- (1) 0.4
- (2) 0.5
- (3) 0.6
- (4) 0.7



91. In a species, the weight of newborn ranges from 2 to 5 Kg 97%. of the newborn with an average weight between 3 to 3.3 kg survive whereas 99% of the infants born with weight from 2 to 2.5 kg or 4.5 to 5kg die.

Which type of selection process is taking place?

- (1) Directional Selection
- (2) Stabilizing Selection
- (3) Disruptive Selection
- (4) Cyclical Selection



92. The frequency of two alleles in a gene pool is 0.19 (1) and 0.81(1). Assume that the population is in Hardy-Weinberg equilibrium. The percentage of heterozygous individuals in the population is

- (1) 0.81
- (2) 0.31
- (3) 0.19
- (4) 1



93. About how many years ago Drypithecus and Ramapithecus were existed?

- (1) 15 million years ago
- (2) 15 billion years ago
- (3) 15 thousand years ago
- (4) 18,000 years ago



94. The Hardy-Weinberg law of equilibrium was based on the following

- (1) Random mating, natural selection, gene flow
- (2) Random mating, genetic drift, gene flow
- (3) Non-random mating, mutation, gene flow
- (4) Random mating, no mutation, no gene flow



95. About _____ the first cellular forms of life appeared on earth.

- (1) 3000 (mya)
- (2) 2000 (mya)
- (3) 2000 (bya)
- (4) 3000 (bya)



96. All organisms share the same genetic code. This statement is in evidence that

- (1) All the organisms looks similar
- (2) The evolution which has occurred was very fast
- (3) The evolution occurs gradually
- (4) All the organisms are descended from a common ancestor



97. From the original seed-eating features, many other forms with altered beaks arose, enabling them to become

- (1) Insectivorous
- (2) Vegetarian finches
- (3) Tortoise
- (4) Both 1 and 2



98. Randomly mating population has an established frequency of for organisms that are homozygous recessive for a given trait. The frequency of this recessive allele in the gene pool is

- (1) 6
- (2) 36
- (3) 0.6
- (4) 16



99. Which is incorrectly matched?

- (1) Jawless fish probably evolved around 350 mya.
- (2) 320 mya - Jawless fishes evolved
- (3) 2000 mya - First cellular form of life appeared
- (4) 65 mya - Dinosaurs disappeared



100. The most accepted line of descent in human evolution is

- (1) Australopithecus → Ramapithecus → Homo sapiens → Homo habilis
- (2) Homo erectus → Homo habilis → Homo sapiens
- (3) Ramapithecus → Homohabilis → Homo erectus → Homo sapiens
- (4) Australopithecus → Ramapithecus → Homo erectus → Homo habilis → Homo sapiens



THANK YOU



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